Team Name: Drishti-AI
Project / Idea Title: Drishti-AI: Explainable DR Screening & Tele-Ophthalmology
Track: MedTech / BioTech / HealthTech • SIH26038 (MathWorks)
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The Problem - Why Does It Matter?

01 PROBLEM STATEMENT

- 77M+ Indian adults live with diabetes; DR is a leading cause of preventable blindness.
- Rural India has ~1 ophthalmologist per 100,000 people - universal specialist screening is impossible.
- 25-35% of field captures are ungradable, yet standard AI still forces a DR grade on them.

02 WHO IS AFFECTED?

- Rural diabetic patients screened at PHCs, sub-centres and mobile vision vans.
- ASHA/ANM workers operating ₹15,000 handheld cameras with no specialist on site.
- District ophthalmologists and health systems absorbing every unnecessary referral.

03 CURRENT PAIN POINTS

- "Garbage-in, garbage-out": blurry images receive confident false negatives or false positives.
- Cloud-only inference needs 5-15 MB per eye over 2G/3G and fails during power or network outages.
- Black-box labels give no visual evidence, quality reason or ABDM/FHIR interoperability.

Idea / Solution

01 WHAT WE'RE BUILDING

- Drishti-AI: an offline-first, dual-stage DR screening and district tele-ophthalmology platform.
- Flutter app for ASHA workers + FastAPI edge server + bilingual PDF and FHIR R4 reports.
- Runs on commodity laptops and ₹15,000 handheld fundus cameras - no GPU, no cloud.

02 HOW IT SOLVES IT

- Model 1 gates domain and image quality; only reliable, unaltered images reach Model 2.
- Model 2 grades 5 DR classes; confidence calibration and Grad-CAM++ explain every release.
- Ungradable, uncertain or high-risk cases route to bounded recapture or human review.

03 CORE VALUE PROPOSTION

- Zero diagnostic leakage: 0% forced predictions on ungradable images vs 100% baseline.
- ~1 s offline inference, 99.1% less telemetry and 73% less specialist review time.
- Clinician-in-the-loop by design: a triage assist that never claims autonomous diagnosis.

How It Works

01 END-TO-END WORKFLOW

- 11-node router: clinical risk / HbA1c → capture → domain + quality gate → grade → confidence.
- GOOD passes; BORDERLINE is reassessed; BAD triggers recapture (hard cap: 2 attempts).
- Low confidence or Grade 3/4 → ophthalmologist review → PDF + FHIR R4 report.

02 USER JOURNEY

- ASHA records risk factors and captures both eyes with a handheld fundus camera.
- On-screen guidance prompts recapture for blur, glare, small pupil or poor field of view.
- ~1.2 s local result; red/amber/green referral slip; only flagged cases sync to the district.

03 TECHNOLOGY STACK

- Flutter (Android/iOS/Web/Windows/macOS), FastAPI, SQLite + SQLCipher offline queue.
- TensorFlow/Keras EfficientNetB0, OpenCV, scikit-learn, Grad-CAM++, MATLAB/Simulink.
- HL7 FHIR R4, SNOMED CT, LOINC, DICOM, Docker deployment and ABDM-aligned workflows.

What Makes Us Different?

01 CORE FEATURES

- Dual quality + domain gate, 5-class DR grading, confidence calibration and abstention.
- Optic disc, fovea and vessel segmentation, MA/exudate candidates and CSME risk distance.
- Grad-CAM++ heatmaps, bilingual PDF reports, audit trail and offline-first sync.

02 EXISTING APPROACH

- Google ARDA / IDx-DR-class systems assume curated hospital images and cloud GPUs.
- They force a prediction on every capture; image quality is an afterthought.
- Cloud upload and black-box output break down on 2G/3G and non-specialist operators.

03 OUR INNOVATION / USP

- Quality-first abstention + bounded recapture + two-tier human review: a safety architecture.
- Non-destructive pixels: no CLAHE, sharpening or generative enhancement before grading.
- Edge-only, hardware-agnostic, ABDM-ready; district-scale proof in MATLAB/Simulink.

Impact & Execution

01 EXPECTED IMPACT

- 0% forced predictions on ungradable images (150-image benchmark); every release is explainable.
- ~1 s on-site turnaround vs ~1.9 min cloud; 99.1% less bandwidth (439.9 → 3.8 GB).
- 73% specialist time saved; 3.7 → 1 tele-reviewer per 100,000 patients per year.

02 BENEFICIARIES

- Rural diabetic patients: earlier triage, fewer unnecessary referrals and repeat visits.
- ASHA/ANM workers: plain-language capture guidance and instant quality feedback.
- District ophthalmologists, PHC administrators and state ABDM programmes.

03 SCALABILITY

- Simulated at 100,000 patients/year across 20 PHCs + 5 mobile vans + 1 district hospital.
- Stateless Docker + FastAPI; camera-specific threshold presets; Android-first client.
- No cloud lock-in; English + Hindi today, Tamil/Telugu/Kannada on the roadmap.

04 IMPLEMENTATION ROADMAP

- Months 0-3: 2-PHC pilot; validate thresholds, workflow and DPDP consent.
- Months 3-6: scale to 20 PHCs + 5 vans; ABDM/FHIR integration and worker training.
- Months 6-12: state rollout, CDSCO Class B/C pathway and outcome audit.

Thank You

Drishti-AI • SIH 2026 • Problem Statement SIH26038 (MathWorks)

Live demo & source: github.com/BezaleelPaul/Drishti-AI